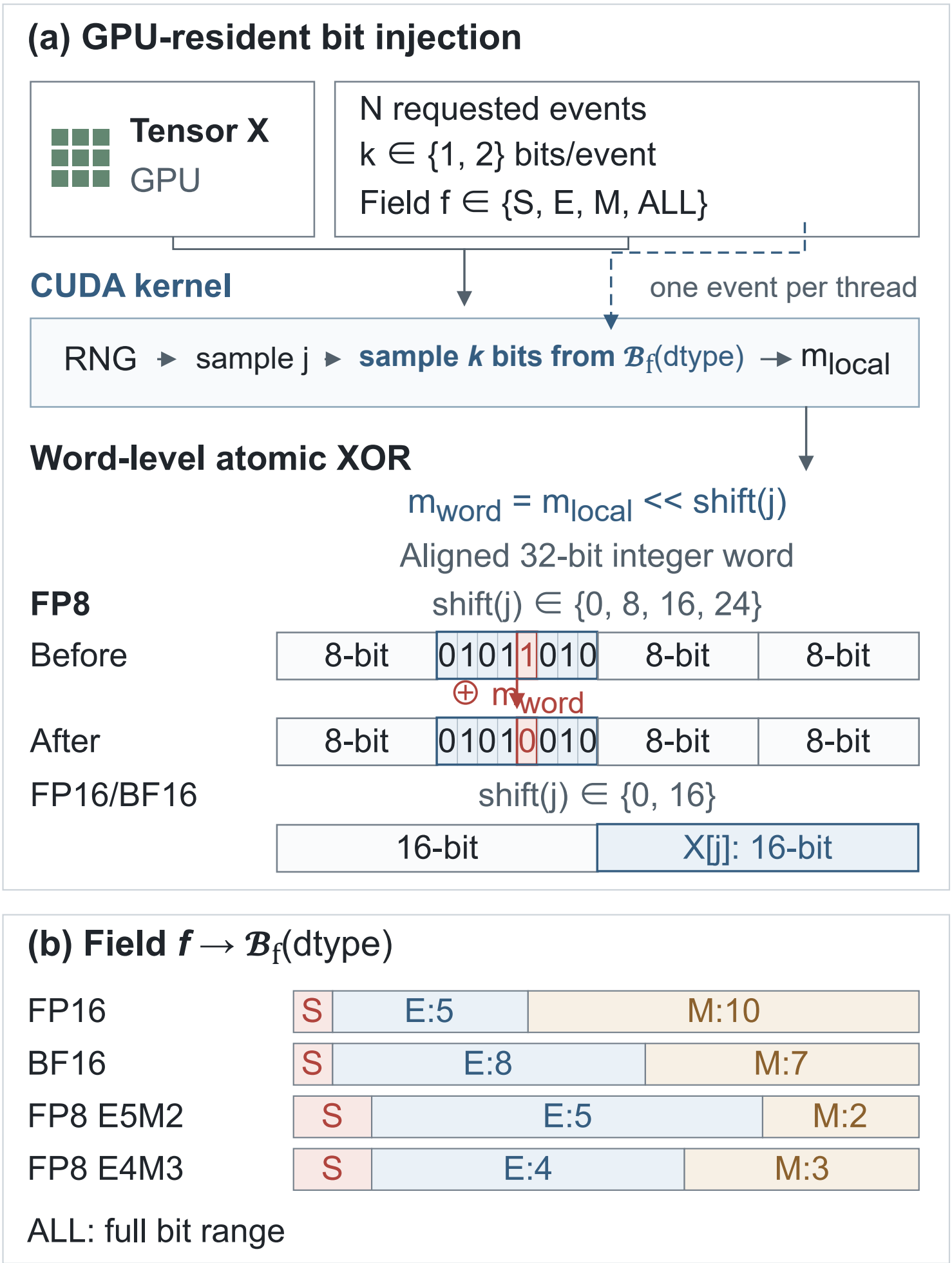

(a) GPU-resident bit injection
Tensor X
GPU
N requested events
k ∈ {1, 2} bits/event
Field f ∈ {S, E, M, ALL}
CUDA kernel
one event per thread
RNG
sample j
sample k bits from $\mathcal{B}_f$(dtype)
m_{local}
Word-level atomic XOR
$m_{word} = m_{local} << shift(j)$
Aligned 32-bit integer word
FP8
shift(j) ∈ {0, 8, 16, 24}
Before
8-bit
01011010
8-bit
8-bit
⊕ m_{word}
After
8-bit
01010010
8-bit
8-bit
FP16/BF16
shift(j) ∈ {0, 16}
16-bit
X[j]: 16-bit
(b) Field f → $\mathcal{B}_f$(dtype)
FP16
S
E:5
M:10
BF16
S
E:8
M:7
FP8 E5M2
S
E:5
M:2
FP8 E4M3
S
E:4
M:3
ALL: full bit range